



Siriyak CR

SENIOR STATISTICAL ANALYST BIOSTATISTICS

Bangalore BTM

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Clinical Data Analyst-Statistician (8+ years total experience, 6+ years in Clinical Domain). Proficient in R & SAS, with expertise in end-to-end data programming for clinical trials, real-world evidence, and analytics, delivering standardized reports and innovative solutions utilizing real-world data.

Skills

Analytical Skills: Generative AI, Advanced Data Analytics, Statistics

LLM: PROMPT ENGINEERING, BERT, GPT, LLMA2, ML: SVM, APRIORI, KNN, NEURAL NETWORK, DATA MINING, RANDOM FOREST, XGBOOSTER, CLASSIFICATION, DECISION TREE, PROBABILISTIC MODELING : BAYESIAN STATISTICS & COMPUTATIONAL STATISTICS, MCMC, PREDICTIVE ANALYTICS, STATISTICAL QUALITY CONTROL, EXACT TEST, REGRESSION MODELING, INFERENCE STATISTICS, SURVIVAL ANALYSIS: COX, KM-(RWPFS, RWOS, RWDOT, RWTTD, TTNT, DOR), HAZARD MODEL, RELIABILITY, DOE, FACTORIAL DESIGN, PCA, FACTOR ANALYSIS, DISCRIMINANT ANALYSIS, MDS, PARTIAL LEAST SQUARES, DIRECT MARKETING, CLUSTER ANALYSIS, MULTIVARIATE ANALYSIS, PATH ANALYSIS, EXPONENTIAL DISTRIBUTION, SAMPLING METHODS

Programming Languages

R, SAS, BASIC-PYTHON & SQL & BASH, GIT, RSTUDIO, COLAB, DBEAVER, BASIC CSS, HTML, & SASS, RMARKDOWN, RESTFUL & PLUMBER APIS,

Visualization-Spotfire, PowerBI, Rshiny

POWER APP, POWER AUTOMATE, EXCEL, POWERPOINT, SHINYDASHBOARD, SHINYTHEMES, SHINYJS, SHINYTEST, SHINYAPPS.IO, WORD, SHARE POINT

Management tools

JIRA, CONFULENCE, WORKDAY

Education

University of Mysore

M.S.C IN STATISTICS

Mysore

Yuvaraj College , University of Mysore

B.S.C IN MATHEMATICS, STATISTICS, COMPUTER SCIENCE

Mysore

Employment

Lead Clinical Database Programmer

KLINERA

Mumbai

April 2024 -> present

- LANGUAGES/TOOLS USED – Veeva, CQL, R, PostgreSQL
- Utilized Veeva for listing and edit checks, study deployment.
- Developed and implemented R scripts for data analysis, reporting, and data quality checks.
- Managed PostgreSQL databases for JReview listings and reports.
- Ensured data quality and integrity through rigorous validation processes.
- Collaborated with cross-functional teams to deliver comprehensive clinical data solutions.
- Provided training and support to team members on data management and analysis tools.
- Analyzed clinical trial datasets with R and SAS to ensure regulatory compliance.
- Created statistical analysis plans (SAP) and supported SDTM/ADaM dataset creation.
- Developed data validation checks and automated listings for safety and efficacy using R and Python.
- Conducted oncology-focused data analysis and authored reports for clinical publications.
- Developed automated algorithms for RECIST 1.1-based tumor response evaluation in R.
- Safety Data Validation: Designed automated checks for AE, CM, and Lab listings using R.
- Statistical Reporting: Created TLFs (Tables, Listings, and Figures) for Phase II Oncology studies.
- validation-Generated a randomization list (independent statistician) and supplied sealed envelopes to the sites for blind-breaking procedures.

Senior Statistical Analyst Biostatistics

Bangalore

CONCERTAI

12 June 2023 → 5 April 2024

- LANGUAGES/TOOLS USED – SAS, RMarkdown, R, Python, Bash, Llama2, GPT3.5, Colab, DBeaver, ChatGPT, Bard, Git, Jira, RShiny, RSconnect
- Led a biostatistics team of two members, overseeing statistical programming tasks, conducting daily scrum meetings for project updates, and conducting code review sessions to ensure accuracy and quality in clinical research analyses
- Automated processes to enhance efficiency
- Assessed patient feasibility for clinical trials
- Conducted data analysis and wrote abstracts
- Performed literature reviews
- Developed sample size determination and statistical design
- Generated analytics tables (SAP) and shells for Phase IV trials, including survival, KM table, and Cox regression
- Defined cohorts: Structured Cohort: A group of patients meeting specific criteria or treated following a structured protocol. Curated Cohort: A subset of patients selected based on specific characteristics relevant for analysis.
- Collected relevant data on patient demographics, treatment regimens, disease characteristics, and survival outcomes from clinical trials, observational studies, or real-world databases.
- Cleaned and prepared the data for analysis, addressing missing values, outliers, and ensuring data quality.
- Performed statistical tests (e.g., log-rank tests, Cox regression analysis) to evaluate differences in survival between cohorts.
- Created visualizations (e.g., Kaplan-Meier curves, survival probability plots) to communicate results effectively. Prepared a clear summary of findings (SAP, TLF), including any statistically significant differences between cohorts.
- Discussed implications of findings in clinical practice and research. Highlighted study limitations and areas for further investigation. Concluded with key takeaways and recommendations based on results.
- Documented methodology, analyses, and results in a comprehensive report or manuscript. Ensured transparency and reproducibility of analysis by providing detailed documentation of the analytical process.
- Assessed rates of medical events from the first anti-cancer drug administration date through study end. Utilized SAS/SQL datasets for data cleaning, preparation, and creation of analysis datasets. Conducted initial statistical review to identify and resolve anomalous values. Analyzed real-world overall survival and progression-free survival as continuous censored outcomes with Kaplan-Meier analysis, specifying rates of survival at specific time points.
- Developed classification models using OpenAI-based tools in R, Python, BASH, Local Machine, and DALL-E for image generation
- Created transcriptions (converted speech to text) using OpenAI's Whisper model in R

Associate Manager Clinical Data Analyst

Bangalore

ELI LILLY AND COMPANY

14 October 2021 → 9 June 2023

- LANGUAGES/TOOLS USED – CDB-Listing, JReview, Veeva EDC-Vault, SAS, RMarkdown, Git, RShiny, Power BI, Power App, RSconnect, ChatGPT, Bard
- Therapeutic Area - Neuroscience, Autoimmune
- Training and Leadership: Trained individuals in portfolio-related tasks, enabling them to lead studies and provide crucial support during the initial phases of clinical trials, ensuring efficient and accurate analysis.
- Automation and Visualization: Implemented automation and visualization techniques, including test data reporting dashboarding with Power BI, to enhance analytics for clinical trials. Increased reusability of statistical reports, ensuring seamless data delivery with real-time validation.
- Patient Prediction: Developed/deployed patient prediction models at early phases of trials to optimize study outcomes.
- Outlier Detection Methods: Developed and deployed outlier detection methods for flagging outlier sites, ensuring data quality and integrity.
- Bayesian Modeling: Developed/deployed Bayesian models to report risk (RBM) in data reports and analytic solutions, providing valuable insights for decision-making.
- Design Study Protocol/SAP: Based study protocols, providing insights into study-level deliverables, data analytics plans, and project plans.
- Collaboration with Data Management and Strategy: Partnered with clinical data management associates and data strategists to deliver data reporting and analytics per business needs. Utilized R, CQL, and SAS to adhere to CRF standards, inform terminologies, and leverage Veeva EDC data structure.
- Study Closeout Support: Developed clean patient trackers during the study closeout phase, ensuring accurate and efficient study completion.
- Regulatory Compliance: Supported submission, inspection, and regulatory response activities by effectively applying knowledge of applicable internal, external, and regulatory requirements, including MQA, CSQ, MHRA, FDA, ICH GCP, PhRMA Privacy knowledge, etc., to all deliverables.
- Automation of TLF Generation: Automated the generation of TLFs based on ADaM/SDTM for trials using R

Clinical Data Analyst / Risk Analyst / Centralized Monitoring

Bangalore

IQVIA

17 Feb 2020 → 13 Oct 2021

- LANGUAGES/TOOLS USED – R, RSHINY, JReview, Rstudio, SpotFire, Veeva EDC, MediRAVE, CTMS, RMarkdown, MSOffice, Git, EXCEL, ETMF
- Therapeutic Area - Autoimmune, Oncology, NeuroScience
- Training individuals in dashboard creation (Spotfire) and automation of portfolio-related tasks enhances efficiency and data visualization capabilities, optimizing decision-making processes in clinical research. by Using R shiny
- A good understanding of drug development processes and strategies with proven ability to apply innovative statistical thinking to meet project or business objectives
- Experience in submissions and interacting with regulatory bodies ICH GCP-Fundamental GCP Exam
- Experience of working with and co-coordinating CROs and Sponsors and building data analytic plan and KRI
- Experience with modeling and simulation and other innovative methodologies using R
- *Used bayesian modeling-Cook distance-Mahalanobis distance-KNN to flag outliers in the site using R
- Providing statistical input to the design analysis reporting and interpretation of clinical studies and contribution to publications
- Identifying developing and implementing novel statistical methodologies in support of IP discontinuation and storage using R shiny
- Developed a dashboard to compare Non-RBM vs RBM based on Key Risk Indicators (KRIs) for investors and sponsors, enhancing transparency and effectiveness of new methods.

Technical consultant Statistician Data analyst

Bangalore

SPSS SOUTH ASIA PRIVATE LIMITED

5 June 2017 -> 14 Feb 2020

- LANGUAGES/TOOLS USED — Basic tableau, R, Rshiny, Git, basic python, IBM SPSS-Statistics-Modeler-Amos-Text analytics for Survey
- Used data visualization statistical analysis pattern recognition, and machine learning
- Along with domain knowledge and subject specific models to solve science engineering and commercial problems
- Assisted the formulation of new projects or Model optimization
- Contribute to all stages of data analytics or decision modeling development deployment and support
- Database historian development data extraction transformation data filtering and quality control
- Processed unstructured data into a form suitable for analysis develop technical reports and presentations
- Created tutorials documentation training material and case studies
- Built simulation models using proprietary and vendor tools to model business problems
- Supervised the Predictive Analytics team of developers during an ongoing project to write code for time series algorithms and survival analysis. validated the output by integrating it with R programs

Pre sales Post sales Technical Associate Statistician

Bangalore

STARCOM INFORMATION TECHNOLOGY LIMITED

16 May 2016 -> 28 April 2017

- LANGUAGES/TOOLS USED — Basic tableau, R, Rshiny, Git, basic python, SPSS Statistics, SYSTAT, Sigmaplot, Sigmastat, Table Curve 2D, Table Curve 3D, Peak fit
- Creating Business presentations Technical documents Case studies and other collateral's for respective Verticals
- Maintaining excellent relations with key clients business groups to generate referrals as well as build avenues for additional business Planning and Implementing Promotional Activities to create Brand Visibility and Market research
- Streamlined training processes through the creation of quick reference to clients work efficiency prospecting
- Toxology research and report generation
- Building automated time series model based on business domain
- Supervised the StarBI team of developers during ongoing project for writing time series algorithm R integration and validated the output
- Used Statistical techniques for hypothesis testing to validate data and interpretations
- Performed market analysis to efficiently achieve objectives increasing sales
- Successfully interpreted data to draw conclusions for managerial action and strategy Presented findings and data to team to improve strategies and operations